

Chelsea Dole

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[Github](#) ▪ [LinkedIn](#) ▪ [Website](#)

TECHNICAL SKILLS

- Python
- Golang
- JavaScript
- SQL
- PostgreSQL
- Apache Kafka
- DynamoDB
- Kubernetes
- Cloud computing (AWS, GCP)
- Docker
- Linux

WORK EXPERIENCE

Citadel, New York City

Database Engineer

July 2024–Present

Building database infrastructure, providing organization-level guidance about best practices, and optimizing performance for all teams at Citadel.

Technologies: PostgreSQL, Python, Shell, Terraform, Kubernetes, GCP, Linux

Brex, Remote (Seattle)

Staff Software Engineer (Tech Lead, Data Storage)

Aug 2023–June 2024

Senior Software Engineer

Feb 2022–Aug 2023

- Optimized read/write performance for PostgreSQL and Redis through query optimization, schema redesign, hash/range partitioning, configuration tuning, automation, and more.
- Contributed to planning & execution of widespread infrastructure cost saving initiatives resulting in 30% and 50% reduction in AWS RDS spend across production and staging respectively.
- Reduced a 7TB PostgreSQL database to 800GB through schema redesign and range partitioning with <100ms downtime, resulting in \$38K/year savings and 60% drop in read latency.
- Pitched, architected, and delivered “DB Metadata Service”, a DynamoDB/Golang-backed gRPC service enabling realtime software/hardware upgrades without impact on metrics or dependencies.

Technologies: PostgreSQL, DynamoDB, Redis, Golang, Terraform, PGBouncer, Kubernetes, AWS, Linux

Coffee Meets Bagel, Remote

Senior Data Engineer (Tech Lead)

May 2021–Present

Data Engineer III

April 2020–May 2021

Data Engineer II

June 2019–April 2020

- Architected and built distributed systems to support ML models and business-critical data warehouses
- Designed and fully implemented an event-based feature engineering system providing real-time features to machine learning models. Resulted in an 85% reduction in feature calculation time and estimated 75%+ reduction in engineering time needed to author features.
- Led project to shard a business-critical 3TB Postgres DB with ~4B writes/day using Citus. Built multiple POC sharding options, performed stress testing, and estimated costs.
- Planned & implemented a company-wide migration from RabbitMQ/Kinesis to Apache Kafka.

Technologies: Python, PostgreSQL, Kafka, Elasticsearch, AWS, Django, Kubernetes, Redis, Docker, Linux

Zonar Systems, Seattle

Software Development Engineer I

Dec 2017–June 2019

Built backend applications to support a data analysis web platform, including components of a Python data analysis API, a Golang event logging API, and a Dataflow pipeline.

Technologies: Python, Golang, JavaScript/Node.js, Kubernetes, GCP, Pandas/NumPy, PostgreSQL

SPEAKING

Postgres Platform “Best Practices” for the Modern DBA ([slides](#)) 2024

50-minute talk about Postgres platform-building. Given at: PGConf NYC 2024, PGConf EU 2024

It's Not You, It's Me: Breaking Up with Massive Tables via Partitioning ([slides](#), [video](#)) 2023

50-minute talk about how to partition live, production-critical PostgreSQL tables while minimizing downtime. Given at: PGConf NYC 2023, PGConf EU 2023, and the Seattle Postgres meetup.

Postgres at Scale: Vacuum, MVCC, & Table Bloat ([slides](#), [video](#)) 2023

30-50 minute talk about understanding, quantifying, and mitigating PostgreSQL “table bloat”. Given at: SoCal Linux Expo 2023, PGDay Paris 2023, Nordic PGDay 2023, PASS Data Summit 2023, etc.

Understanding Python's Memory Management ([slides](#), [video](#)) 2021

Short introduction to Python memory management. Given at: PyLadies Int'l Women's Day 2021.

Dreaming of Databases: 5 Questions for Data Layer Decisions ([slides](#)) 2020

25-minute talk about database vocabulary and architecture. Given at: ThaiPy Bangkok.

Going Serverless: Deploying with GCP App Engine & Cloud Functions ([slides](#)) 2019

30-minute talk about API deployment with App Engine. Given at: Puget Sound Python meetup.

PROJECTS // OPEN SOURCE

[PostgresUS Foundation](#), Chair (Diversity Committee) Mar 2023–Present

Responsible for allocation and management \$10K annual funding with the goal of supporting PostgreSQL community members from underrepresented backgrounds.

[Invest in Chia](#), Co-Founder 2021–2022

Technical co-founder of a small “green” cryptocurrency startup in the Chia coin space. Generated and transferred 11PB+ of data to customers across 500+ bare metal servers.

EDUCATION

Code Fellows, Seattle, WA 2017

Certificate: Backend Software Development in Python

University of California, Berkeley 2013–2017

Degree: Political Economy (B.A.)

GPA: 3.40